

**CODE:**

```
clear all
N = 10^6; % number of bits or symbols rand('state',100);
% initializing the rand() function randn('state',200);
% initializing the randn() function

% Transmitter
ip = rand(1,N)>0.5; % generating 0,1 with equal probability
s = 2*ip-1; % BPSK modulation 0 -> -1; 1 -> 1
n = 1/sqrt(2)*[randn(1,N) + j*randn(1,N)]; % white gaussian noise, 0dB variance
Eb_N0_dB = [-3:10]; % multiple Eb/N0 values

for ii = 1:length(Eb_N0_dB)
    % noise addition
    y = s + 10^(-Eb_N0_dB(ii)/20)*n; % additive white gaussian noise

    % receiver - hard decision decoding
    ipHat = real(y)>0;

    % counting the errors
    nErr(ii) = size(find([ip - ipHat]),2);
end

simBer = nErr/N; % simulated ber
theoryBer = 0.5*erfc(sqrt(10.^(Eb_N0_dB/10))); % theoretical ber

% plot
close all
figure
semilogy(Eb_N0_dB,theoryBer,'b.-');
hold on
semilogy(Eb_N0_dB,simBer,'mx-');
axis([-3 10 10^-5 0.5])
grid on
legend('theory','simulation');
xlabel('Eb/No, dB');
ylabel('Bit Error Rate');
title('Bit error probability curve for BPSK modulation');
```

## RESULT:

